STORAGE



Vertical Optimizations for KV$ Infrastructure

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SAMSUNG

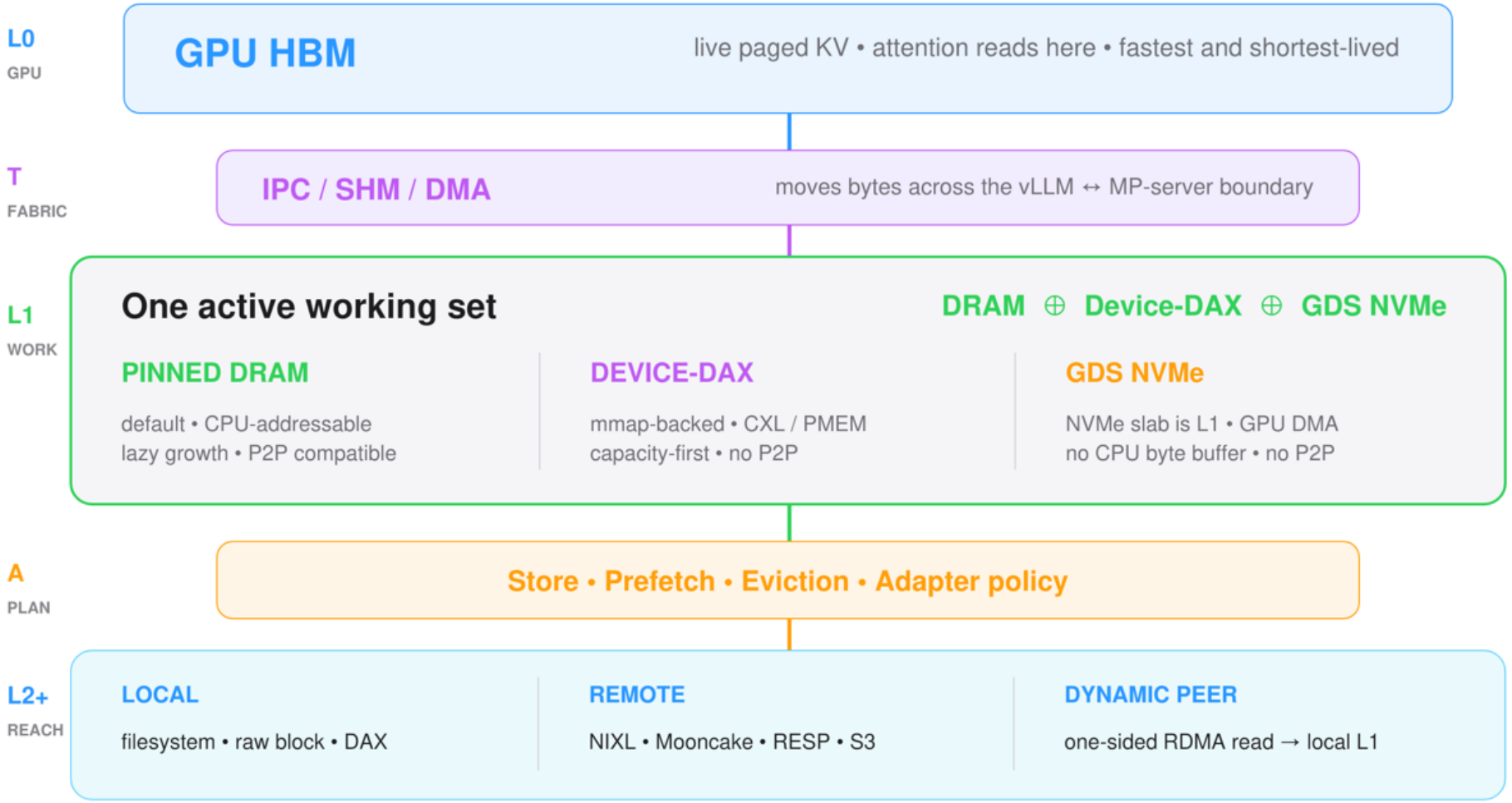
The Era of KV$

- Significant growth of KV$ projects
- Convergence into a few *mainline* KV$ projects
  - LMCache, Mooncake, HiCache
- Good integration in end-user systems
  - vLLM, SGLang, TRT-LLM …
- OCP Storage provides a forum for common SSD requirements
  - Datacenter NVMe SSD work includes Flexible Data Placement (FDP)

KV$: More than a workload

- KV$ is a new type of AI-native data
- KV$ is ephemeral
- KV$ can be recomputed
  - Compare DRAM capacity, SSD retrieval and recompute cost
- KV$ retention depends on workload reuse and cache policy
  - Measure reuse and retention across workload classes

LMCache: Storage Architecture – 1. HW view



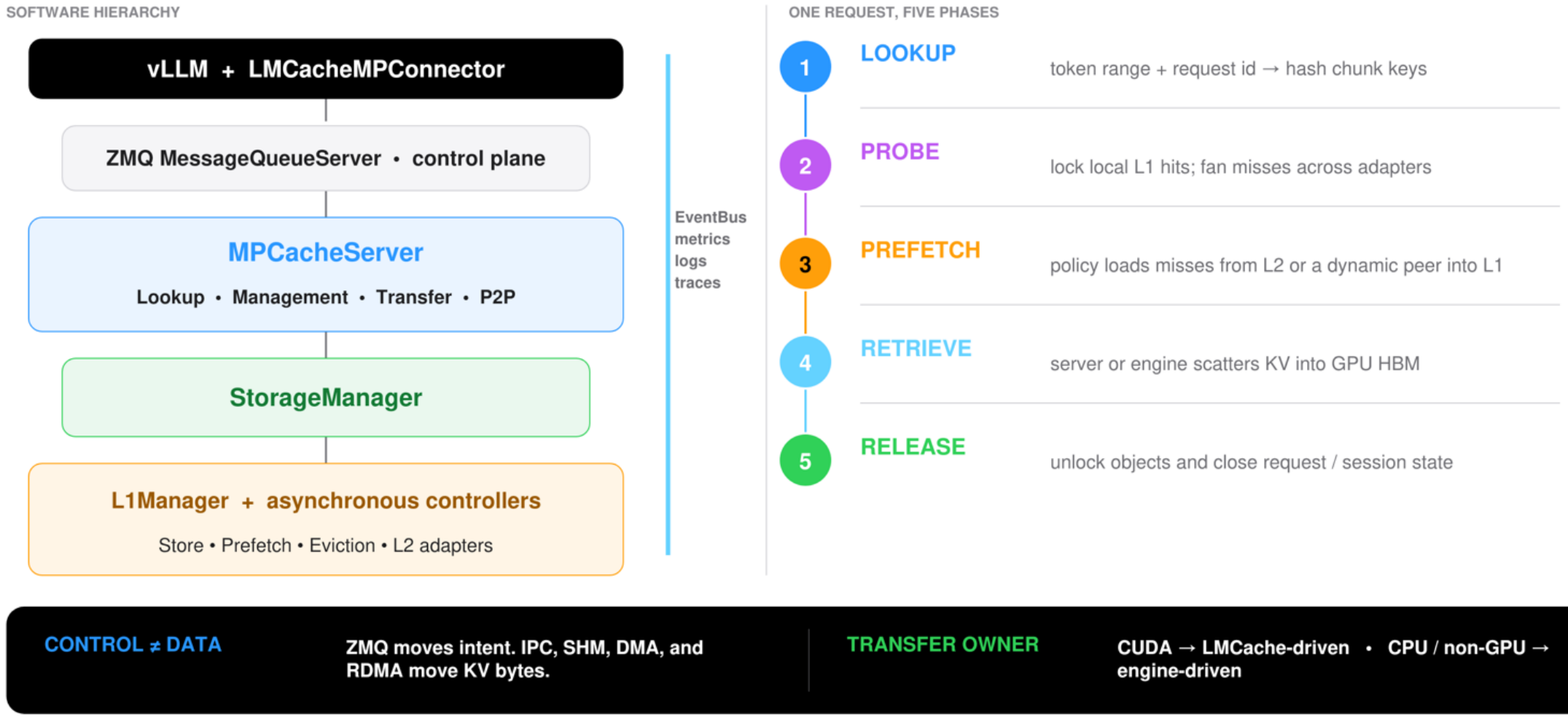
Three laws

1 **HBM is always the live execution home.**

2 **L1 is one of DRAM, DAX, or GDS — never all three.**

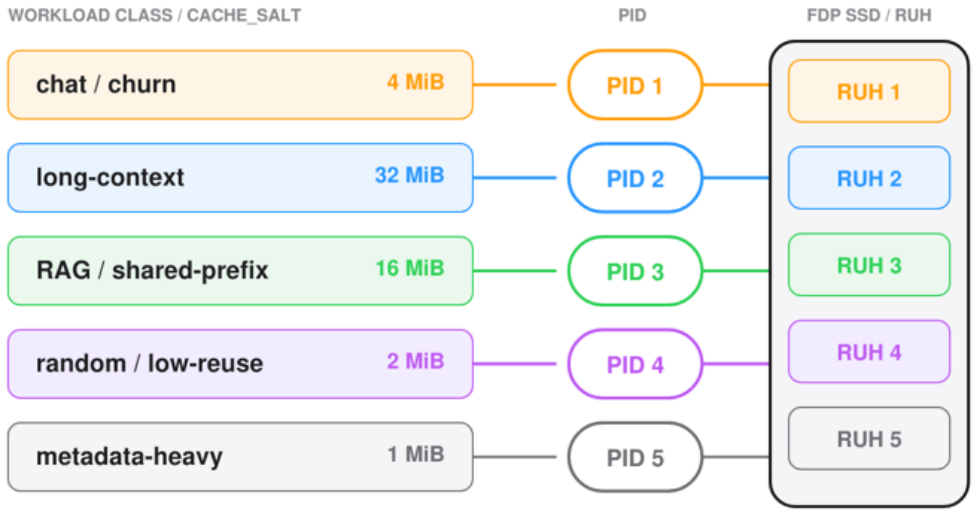
3 **P2P requires one contiguous pinned-DRAM L1.**

LMCache: Storage Architecture – 2. SW view



LMCache: Optimization - Data placement

MECHANISM

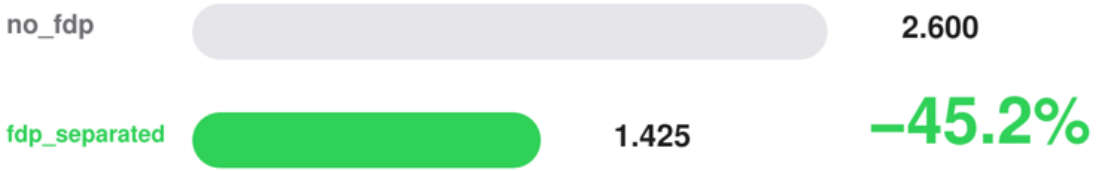


Like lifetimes reclaim together.

Less cross-stream garbage collection and write movement.

MEASURED EFFECT

WRITE AMPLIFICATION FACTOR



AVERAGE WRITE

–55.6%

25,364.155 → 11,249.847 ms

AVERAGE READ

–35.7%

36.375 → 23.386 ms

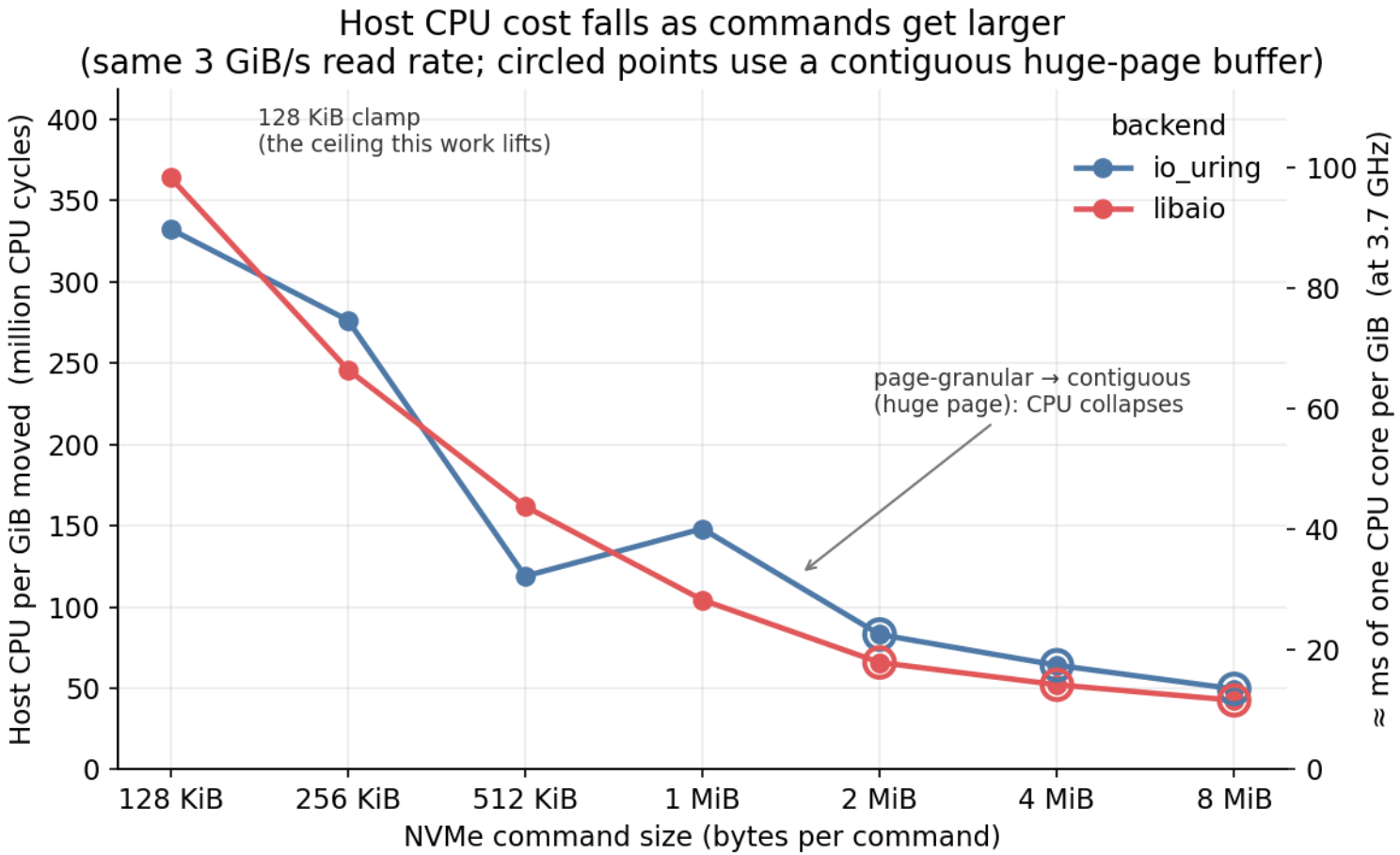
p90 latency: write –29.4% • read –22.0%

PR TRACE-REPLAY CONDITIONS

PM9D3a	960 GiB	89%	4×	9 workers
FDP SSD	configured	utilized	capacity writes	855 GiB trace

Synthetic storage-level LMCache traces; sequential fio prefill before each run. Latency is end-to-end replayed object-I/O completion—not serving TTFT. Residual WAF may reflect the device's Initially Isolated policy and 16-GiB reclaim units.

LMCache: Optimization - Large Transfers



- CPU cost falls from ~473–592 to ~38–57 Mcyc/GiB (10–12×).
- Simulation predicts ~23 Mcyc/GiB. Beyond 4 MiB, additional savings diminish.
- OCP input: transfer-size and buffer-layout trade-offs.

Proposed contribution to OCP Storage

- Share reproducible KV-cache workloads and test configurations
- Evaluate write amplification, tail latency, throughput and CPU cost
- Bring workload evidence to OCP discussions on SSDs for AI inference

Proposed scope for community discussion
opencompute.org/community/storage

Takeaways

KV$ memory demands is real.

KV$ has unique data characteristics.

We have SW-HW co optimization room in Storage offloading.

Talk to us about further KV$ Optimizations in SSDs

Visit our booth ☺

Thank You!

Questions & Comments

